

AI Intelligent Freight Quote Generation System

Complete M1–M3 Architecture • AI Agents • Dashboards • Process Flow • Test Data

Project Goal

A connected platform where a customer submits shipment information, AI/ML intelligence analyzes pricing and risks, an operations team reviews the result, and a final freight quote is delivered to the customer.

Document Scope

Area	Coverage
M1	Shipment, route and rule-based quote generation
M2	AI/ML freight price prediction
M3	Weather, customs, route and composite risk
Users	Customer, Freight Agent, Customs Officer, Admin
AI	Orchestrator, Route, Pricing, Weather, Customs and Risk Agents

1. Users and Responsibilities

Customer

Creates shipment requests, uploads information/documents, requests quotes, views final quotes and accepts or rejects them.

Freight Agent / Operations

Reviews shipment requests and AI analysis, validates commercial details, modifies when required and sends the final quote.

Customs Officer

Reviews customs requirements and documents, adds flags/remarks and provides customs-related risk information.

Admin

Manages users, permissions, master data, system configuration, AI monitoring, analytics and audit activity.

Role Connection

Customer submits data → AI services analyze it → Freight Agent performs human review → Customs Officer participates when customs review is required → Admin manages and monitors the platform.

2. How M1, M2 and M3 Connect

M1 — Core Quote Foundation

M1 creates the shipment, validates information, determines route details and calculates an initial rule-based price.

Customer → Create Shipment → Validation → Route / Distance / ETA → Rule-Based Pricing → Draft Quote

M2 — AI Pricing Intelligence

M2 receives the shipment and route data created by M1. A trained ML model predicts freight price from historical patterns.

M1 Data → Feature Preparation → ML Model → AI Predicted Price → Compare with Rule Price → Recommended Price

M3 — Risk Intelligence

M3 uses the same shipment context to analyze weather, customs and route risks and produces a composite risk score.

Shipment + Route → Weather Analysis + Customs Analysis + Route Risk → Risk Engine → Overall Risk Score

Key Connection: M1 creates operational data. M2 adds pricing intelligence. M3 adds risk intelligence. The Quote Engine combines all outputs.

3. Complete End-to-End Process

1. Customer Login

Customer signs in and starts a quote request.

2. Create Shipment

Enter origin, destination, cargo, weight, volume, mode, container and documents.

3. Validate & Store

System validates data and creates a Shipment ID.

4. Route Intelligence (M1)

Determine route options, distance and estimated transit time.

5. Rule Pricing (M1)

Calculate base, distance, weight, fuel and handling costs.

6. AI Pricing (M2)

ML model predicts an intelligent freight price.

7. Risk Intelligence (M3)

Weather, customs and route risks are analyzed.

8. Composite Risk

Risk Engine calculates LOW/MEDIUM/HIGH or numeric risk.

9. Quote Engine

Combines shipment, route, prices and risk into a draft quote.

10. Human Review

Freight Agent approves, modifies, requests information or rejects.

11. Final Quote

Approved quote is sent to the customer.

12. Customer Decision

Customer accepts or rejects; lifecycle is updated.

Master Flow

CUSTOMER → SHIPMENT → M1 ROUTE + RULE PRICE → M2 AI PRICE → M3 WEATHER/CUSTOMS/ROUTE RISK → QUOTE ENGINE → AGENT REVIEW → FINAL QUOTE → CUSTOMER ACCEPT/REJECT

4. AI Intelligence Layer

AI Pricing Model

ML regression predicts freight price using historical features such as distance, weight, mode, container, fuel and carrier.

Route Intelligence

Recommends suitable routes using distance, transit time and operational criteria.

Weather Intelligence

Analyzes route-related weather conditions and possible delay risk.

Customs Intelligence

Checks required documents, missing requirements, restrictions and customs risk.

Risk Prediction Engine

Combines multiple signals into one explainable composite risk score.

AI Quote Recommendation

Uses pricing and risk outputs to recommend a commercially sensible quote.

5. AI Agents

Agent	Responsibility	Output
AI Orchestrator	Controls workflow and calls required agents	Combined analysis state
Route Agent	Route options, distance and ETA	Recommended route
Pricing Agent	Rule price + ML prediction comparison	Recommended price
Weather Agent	Weather and delay analysis	Weather risk
Customs Agent	Documents and customs requirements	Customs risk
Risk Agent	Combines all risk signals	Overall risk + recommendation

Agent Flow

AI ORCHESTRATOR → ROUTE AGENT → PRICING AGENT → WEATHER AGENT + CUSTOMS AGENT → RISK AGENT → QUOTE ENGINE

6. Dashboard Architecture

Dashboards are for human users. AI Agents are backend intelligence services. This distinction keeps the architecture clean.

Customer Portal

Side Navigation: Dashboard • My Shipments • Request Quote • My Quotes • Documents • Notifications • Profile

Dashboard Content: Cards: Total Shipments, Active Requests, Pending Quotes, Accepted Quotes, Recent Activity

Freight Agent Portal

Side Navigation: Dashboard • Shipment Requests • All Shipments • Quote Requests • Quote Review • Generated Quotes • AI Pricing Analysis • Risk Analysis • Customers • Documents • Notifications • Profile

Dashboard Content: Cards: New Requests, Pending Reviews, High Risk Shipments, Quotes Sent Today

Customs Officer Portal

Side Navigation: Dashboard • Pending Reviews • Assigned Shipments • Document Verification • Customs Risk Flags • Completed Reviews • Notifications • Profile

Dashboard Content: Cards: Pending Reviews, Missing Documents, High Risk Cargo, Completed Today

Admin Portal

Side Navigation: Dashboard • Users • Customers • Freight Agents • Customs Officers • Roles & Permissions • All Shipments • All Quotes • AI Pricing Monitor • AI Agent Monitor • Risk Intelligence • Locations • Routes • Carriers • Container Types • Cargo Categories • Pricing Rules • Reports • Notifications • Settings • Audit Logs

Dashboard Content: Cards: Total Users, Shipments, Quotes, Pending Reviews, High Risk Alerts, AI Predictions, Analytics

7. How Dashboards Connect to the Workflow

Customer Dashboard

Request Quote → Shipment Created → Processing Status → View Final Quote → Accept/Reject

Freight Agent Dashboard

New Shipment → View Route + AI Price + Risk → Review/Modify → Approve → Send Quote

Customs Dashboard

Assigned Shipment → Verify Documents → Add Remarks → Approve/Flag → Result returns to Risk workflow

Admin Dashboard

Manage Users + Master Data + AI Performance + Operations + Analytics + Audit

8. Example Test Shipment Data

Field	Test Value
Customer	ABC Electronics Pvt Ltd
Shipment ID	SHP-1001
Origin	Chennai, India
Destination	Rotterdam, Netherlands
Cargo Type	Electronics
Weight	5,000 KG
Volume	12 CBM
Transport Mode	Sea
Container	40FT
Distance	8,950 KM
Estimated Transit	24 Days

Example AI & Quote Results

Component	Example Output
Rule-Based Price	■87,000
AI Predicted Price	■85,500
Recommended Price	■86,000
Weather Risk	30/100 — Moderate
Customs Risk	40/100 — Medium
Route Risk	20/100 — Low
Overall Risk	MEDIUM
Quote Status	PENDING REVIEW

9. Core Test Scenarios

#	Scenario	Expected Result
1	Valid shipment submission	Shipment created with SUBMITTED status
2	Required field missing	Validation error
3	Route calculation success	Distance and ETA stored
4	ML prediction success	AI price stored
5	ML service unavailable	Fallback to rule price
6	High weather risk	Risk alert added
7	Missing customs document	Customs flag/action required
8	High composite risk	Stronger human review required
9	Agent modifies price	Reason and audit record stored
10	Agent approves quote	Final quote sent to customer
11	Customer accesses another quote	Access denied
12	Customer accepts quote	Status updated to ACCEPTED

10. Status Flow & Final Architecture

Shipment Status

DRAFT → SUBMITTED → PROCESSING → ANALYZED → QUOTED → CLOSED / CANCELLED

Quote Status

DRAFT → GENERATED → PENDING_REVIEW → APPROVED → SENT → ACCEPTED / REJECTED / EXPIRED

Final Architecture

CUSTOMER PROVIDES DATA → SYSTEM VALIDATES → M1 BUILDS ROUTE + RULE PRICE → M2 PREDICTS AI PRICE → M3 ANALYZES WEATHER/CUSTOMS/ROUTE RISKS → RISK ENGINE SCORES → QUOTE ENGINE COMBINES RESULTS → FREIGHT AGENT REVIEWS → FINAL QUOTE → CUSTOMER DECIDES → ADMIN MONITORS EVERYTHING

Final Summary

This is one connected intelligent freight quotation workflow—not three separate projects. M1 provides the foundation, M2 adds machine-learning price intelligence, and M3 adds operational risk intelligence. AI Agents perform analysis, human dashboards provide control, and the Quote Engine connects all results into one final business decision.